Reg. No. | | | | | | | | | | | [13ME651/12ME651]

SIXTH SEMESTER B.Tech./B.E. EXAMINATION, JUNE – 2016

HEATING, VENTILATION, REFRIGERATION AND AIR CONDITIONING

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data, if any, may be suitably assumed.
3. Use of relevant thermodynamic data hand book, refrigeration charts and psychrometric charts permitted.

1. a. List the different type of heating system. Explain with neat sketch natural circulation warm air heating system. **07**

b. Describe briefly panel heating system. Mention the merits and demerits of panel heating system. **07**

2. a. What is ventilation and why it is required? Write a short note on natural ventilation. **07**

b. Write briefly on: **07**

i. Mechanical ventilation

ii. Types of air outlets

3. a. Define the following: **07**

i. Degree of saturation

ii. Relative humidity

iii. Vapour density

iv. Dew point temperature

v. Wet bulb temperature

vi. Adiabatic saturation template

vii. Enthalpy of moist air.

b. Air flowing at 100kg/min and at DBT of 30°C, WBT of 21°C mixes adiabatically with another stream of air flowing at 150kg/min and at DBT of 21°C, RH of 50%. Using psychrometric chart, determine the specific humidity and enthalpy of the two stream mixed together as a third stream in a steady flow process. **07**

4. a. Mention the sensible and latent heat loads generally considered in cooling load calculation. **04**

b. An air conditioned hall is to be maintained at 27°C DBT and 21°C WBT. It has sensible heat load of 46.5 KW and latent heat load of 17.5 kW. The air supplied from outside atmosphere is at 38°C DBT and at 27°C WBT at $25m^3$/min rate directly into the room. Outside air to be conditioned is passed through cooling coil whose ADP is 15°C. The quantity of recirculated air from the hall is 60%. The quantity of air is mixed with conditioned air after the cooling coil. Determine **10**

i. The condition of air after the coil and before the recirculated air mixes with it

ii. The condition of air entering the hall

iii. The mass of fresh air entering the cooler.

iv. The bypass factor of the cooling coil and

v. The refrigeration load on the cooling coil.

5. Write short notes on the following: **14**

i. Food preservation

ii. Air conditioning of theatre

iii. Air conditioning of Automobiles.

6. a. With a neat schematic diagram explain air refrigeration system working on Bell Coleman cycle. Draw P.V and T.S. diagram of the cycle and obtain the expression for COP in terms of temperature before and after expander. **06**

b. A dense air refrigeration machine works on Bell Coleman cycle with a 10 ton of refrigeration capacity. The cooler pressure is 4.2 bar and refrigeration pressure is 1.4bar. The air is cooled to temperature of 50°C in the cooler. The temperature of air at inlet to compressor is -20°C. Determine for the ideal cycle **08**

i. COP

ii. Mass flow of refrigerant/ton of refrigerant

iii. Piston displacement

iv. Carnot COP (or ideal COP)

v. Heat rejected in condenser.

Take 1 ton of refrigeration = 210 KJ/min and γ=1.4 for air R=287J/kg°C, C_p = 1kJ/kg°C.

7. a. List the properties of refrigerant absorbent combination **03**

b. With T-S and P-h diagram explain the working of simple vapour compension refrigeration system. Also draw a neat shematic diagram of the system. Write an expression for the COP of the system. **07**

c. In a vapour absorption system, heating, cooling and refrigeration takes place at the temperatures of 100°C, 20°C and -5°C respectively. Find the maximum COP of the system. **04**

8. a Write short notes on **08**

i. Manufacture of dry ice

ii. Cascade refrigeration system

b. The temperature limits of Ammonia refrigerating system are 25°C and -10°C. If the gas is dry at the end of compression, calculate the COP of the system, assuming no under cooling of liquid Ammonia takes place. Take the properties of NH_3 as given in table below: **06**

Temperature °C	Liquid heat KJ/kg	Latent heat KJ/kg	Liquid entropy KJ/kgK
25°	298.9	1166.94	1.1242
-10	135.37	1297.68	0.5443